

Springer Texts in Statistics

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An Introduction to Statistical Learning

with Applications in R

 Springer

ANALYTICS

Week 10

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*Examples Inspired By
CAS, SOA, Coaching
Actuaries ©*

Exam #2 Info

- 15 True/False conceptual questions (1×15) + 5 Numerical Free Response questions (5×5 points) = 40 Points
- 75 Minutes
- See Study Guide for Q Distribution – This review won't cover all the topics.

Q1 T/F Practice

- 1. If multiple linear regression error terms are not independent, estimated standard errors are too high.
- 2. Under multiple linear regression, the prediction interval is always at least as wide as the confidence interval.
- 3. Residual standard error can be a good metric for selecting the best model among candidates with different number of predictors.

Q2 T/F Practice

- 4. Under least squares, the leave-one-out-cross-validation estimate of test MSE can be found by fitting one model.
- 5. Larger values of Mallows's C_p indicate models with higher estimated test error.
- 6. Compared to k -fold cross-validation, LOOCV has lower bias and higher tendency to overestimate the test error rate.

Q3 T/F Practice

- 7. The hierarchical principle states that if an interaction term is significant, the individual terms should remain in the model, even if their t-tests are insignificant.
- 8. In multiple linear regression, the F-test for the entire model tests whether at least one regression coefficient (excluding the intercept) is non-zero.
- 9. Multi-collinearity invalidates the reliability of whole-model F-tests.

Q4 T/F Practice

- 10. A high leverage point must have at least one predictor value that is extreme relative to that predictor's distribution.
- 11. If Model I uses 1 predictor and has an R^2 of 0.4, whereas Model II uses 2 predictors and has an adjusted R^2 of 0.42, Model II will always be preferable.
- 12. Adjusted R^2 can be negative for poorly performing models.

Q5 T/F Practice

- 13. LOOCV is equivalent to k-fold cross-validation with k equal to the number of observations.
- 14. A QQ plot with residual points mostly following a straight reference line means the model satisfies homoscedasticity.
- 15. A categorical predictor with four categories should generally be represented by four dummy variables plus an intercept.

Q6

- The following regression model: $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$ is fit to 5 data points.
- $\hat{y} = 10 + 2x_1 - x_2$
- For the second data point, $y_2 = 12, x_{21} = 3, x_{22} = 5$.
- The sum of the other 4 residuals squared, $e_1^2 + e_3^2 + e_4^2 + e_5^2$, is 30.
- $\sum_{i=1}^n (y_i - \bar{y})^2 = 100$
- Calculate the adjusted R^2 for this model to the nearest percent.

Q7 (CAS)

- You fit a multiple regression with 12 continuous explanatory variables and one intercept to predict the # of tweets made by politicians.
- Error sum of squares is 460.1.
- The 10th diagonal entry of the hat matrix, $h_{10,10} = 0.35$
- The 10th residual value $e_{10} = 9.5$
- Cook's Distance $D_{10} = 5.0$
- Calculate the # of data points this model was fit to.

Q8

- A regression model is built off 100 observations: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$.
- Calculate the variance inflation factors and comment on the collinearity.

Model	<i>SSReg</i>	<i>SSE</i>
$y \sim x_1 + x_2 + x_3$	250	150
$x_1 \sim x_2 + x_3$	10	1
$x_2 \sim x_1 + x_3$	20	100
$x_3 \sim x_1 + x_2$	1000	200

Q9

- In a multiple linear regression model,
- The full model fits 10 explanatory variables. Its residual standard error is 9.
- Model #1 uses 4 continuous explanatory variables and one categorical variable that has 3 levels. Its residual sum of squares is 10,000.
- Model #2 uses 4 continuous explanatory variables and one interaction term. Its error sum of squares is 10,373.
- If BIC is virtually indifferent between Models 1 & 2, which model does AIC prefer?

Q10 (SOA)

- You perform a regression of y on x_1 and x_2 with 6 observations. You determine:
- $\hat{y}_i = 20 - 1.5x_{i1} - 2x_{i2}$
- Regression sum of squares = 42
- Residual sum of squares = 12
- The standard error of the b_2 estimate is 1.633.
- A. What is the mean squared error?
- B. Calculate the t -statistic for testing $H_0: \beta_2 = 1$ and smallest listed α where H_0 would be rejected.
- C. If a Model 2 of $Y = \beta_0 + \beta_1 x_1 + \varepsilon$ is created from the same data and the regression sum of squares is 30, calculate the F -statistic that tests the hypothesis: $\beta_2 = 0$.